

TPC-H SF10 Benchmark Report

PostgreSQL 17 · AMD EPYC 9555P · with Power Consumption

Run date: June 09, 2026 15:52 UTC

23,543

QpH@SF10

Composite score

16,797

Power@SF10

Geo mean 2.14s

33,000

Throughput@SF10

4 streams · 96s

87 W

Avg Power (query)

Package-0 during power test

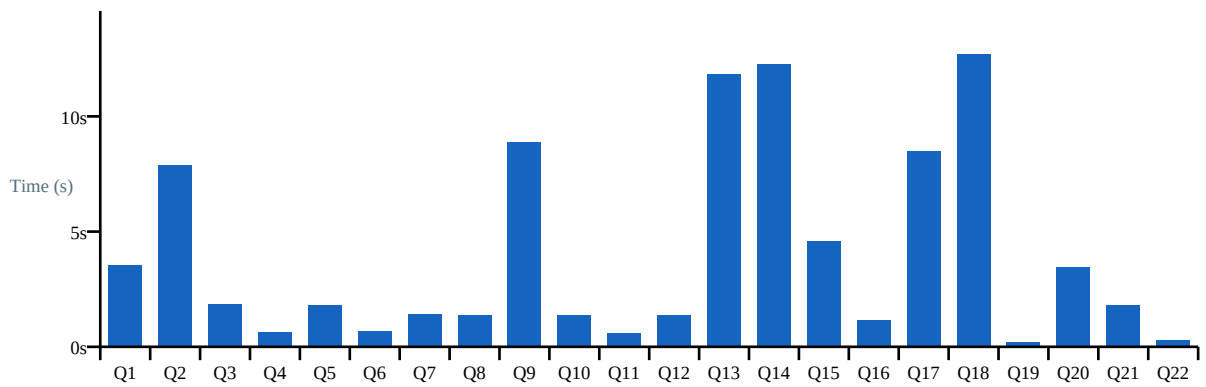
System & Configuration

Component	Detail
CPU	AMD EPYC 9555P — 64 cores / 128 threads, single socket (Turin / Zen 5)
Bench CPUs	32 logical CPUs — NUMA node 0 (CPUs 0-15, 64-79), NPS4
RAM	377 GiB DDR5, 12 memory channels
Storage	465 GB NVMe SSD
OS	Ubuntu 24.04 LTS (kernel 6.8.0-83-generic)
PostgreSQL	17 (Docker image postgres:17)
HammerDB	5.0
Scale factor	SF10 — 10 GB raw data, ~20 GB on disk with indexes

Parameter	Value	Parameter	Value
shared_buffers	150 GB	wal_level	minimal
work_mem	8 GB	autovacuum	off
jit	on	max_parallel_workers	64
Build threads	16	Power test dop	8
Throughput VUs	4	Throughput dop	4

TPC-H Power Test (1 VU, dop=8)

The power test runs all 22 queries sequentially with a single virtual user at degree-of-parallelism 8. The primary metric is **Power@SF10** = SF × 3600 / geometric_mean_query_time.



Per-query execution times. Q13 (11.8s), Q14 (12.2s), and Q18 (12.7s) are the longest — all involve heavy aggregations or large joins.

Query	Time (s)	Query	Time (s)	Query	Time (s)
Q1	3.530	Q2	7.873	Q3	1.827
Q4	0.623	Q5	1.822	Q6	0.667
Q7	1.402	Q8	1.386	Q9	8.882
Q10	1.355	Q11	0.577	Q12	1.351
Q13	11.833	Q14	12.237	Q15	4.565
Q16	1.136	Q17	8.484	Q18	12.674
Q19	0.207	Q20	3.447	Q21	1.796
Q22	0.265				

Total time	Geo mean	Power@SF10
88 s	2.143 s	16,797

TPC-H Throughput Test (4 VUs, dop=4)

The throughput test runs all 22 queries across 4 concurrent streams. The primary metric is **Throughput@SF10** = $(S \times Q \times SF \times 3600) / T_{TT}$, where T_{TT} is the elapsed time of the longest stream.

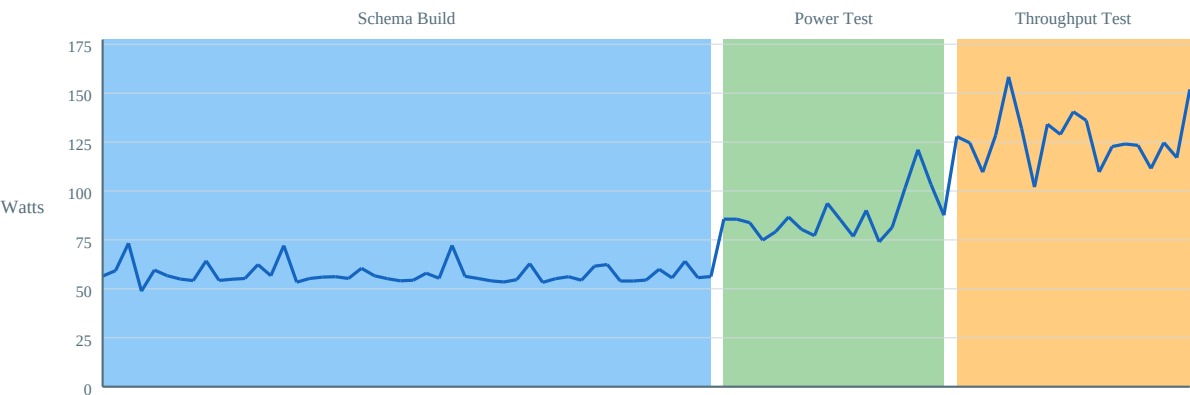
Stream (VU)	Total Time (s)	Geo Mean (s)	vs Longest
Stream 1	96	2.532	baseline (longest)
Stream 2	95	2.504	+1s
Stream 3	94	2.451	+2s
Stream 4	94	2.422	+2s

Streams	Longest stream	Throughput@SF10
4	96 s	33,000

$$\text{QphH@SF10} = \sqrt{(16,797 \times 33,000)} = 23,543$$

Power Consumption (AMD RAPL — Package-0)

Power sampled every 5 seconds via AMD RAPL energy counter (/sys/class/powercap/intel-rapl:0/energy_uj). Measures total CPU package power including cores, uncore, and memory controller. Does not include DRAM or storage device power.



Package-0 power over time. Coloured bands: Schema Build Power Test Throughput Test

Phase	Sample s	Duration	Min W	Avg W	Max W	Avg Energy (Wh)
Schema Build	48	240s (~4m 0s)	49	58	73	3.8
Power Test	18	90s (~1m 30s)	74	87	121	2.2
Throughput Test	19	95s (~1m 35s)	102	127	158	3.3

Energy (Wh) = avg_watts × duration_seconds / 3600. The power test phase represents the actual benchmark-under-test period. Higher parallelism during throughput test drives higher sustained power draw.

Performance per Watt

Metric	Value	Notes
QphH per Watt	220.2	QphH@SF10 ÷ avg(power_test, throughput_test) watts
Power test avg power	87 W	Range: 74–121 W during single-stream queries
Throughput test avg power	127 W	Range: 102–158 W during 4 concurrent streams
Power headroom	153 W below TDP	EPYC 9555P TDP = 280 W; peak measured power well below TDP

Informal benchmark — not TPC-audited. Power measured via AMD RAPL (package-0 energy counter). Source: github.com/skariapaul/postgresql-bench. Report generated 2026-06-09 16:03 UTC.